

TESTs - Lesson 05 - VSCode setup and Code Quality

Why is Hardhat introduced as the new tooling for dApp development, moving beyond Remix?

- A Hardhat is a cloud-based IDE, making local setup unnecessary.
- B It guides developers to use JavaScript/TypeScript for building dApps, including front-end and back-end integration.**
- C It is an older, more stable alternative to Remix for basic smart contract deployment.

What is the main reason to use OpenZeppelin contracts as a reference repository?

- A To copy-paste code directly without understanding it.
- B To leverage community-vetted, secure, and reusable modular definitions for smart contracts like ERC20 and ERC721.**
- C To develop custom, highly experimental smart contract standards.

What tool is recommended for managing multiple Node.js versions and avoiding potential issues?

- A Git CLI
- B NPM (Node Package Manager)
- C NVM (Node Version Manager)**
- D VS Code

What are the key aspects of "code quality" that the tools we're using aim to improve?

- A Code compactness and obfuscation for security.
- B Composability, upgradeability, maintainability, and readability.**
- C Speed of development at the expense of long-term stability.

Describe the process of setting up a new Hardhat project from scratch, including initialization and local installation of Hardhat.



- A First, use `'git clone'` to copy an existing Hardhat project, then modify it. Hardhat is then automatically installed as part of the clone.
- B Create an empty directory, navigate into it, run `'npm init -y'` to create `'package.json'`, then install Hardhat locally using `'npm install --save-dev hardhat'`, and finally initialize Hardhat using `'npx hardhat init'` selecting the TypeScript project option.**
- C Use the `'hardhat create'` command directly from any directory, which automatically sets up everything including Node.js.

How does running tests, as demonstrated with the `'npm run test'` command after modifying the `'decimals()'` function, contribute to code quality and developer confidence?



- A Tests automatically fix bugs in the code without developer intervention.
- B Running tests ensures that any changes to the codebase, even seemingly minor ones, do not introduce unexpected regressions or break existing functionality, thereby validating code correctness and providing 'peace of mind'.**
- C Tests are primarily used to generate documentation for the smart contracts.

What are the two recommended VS Code extensions for Solidity development and what specific benefit does each provide?



- A ESLint (for syntax highlighting) and Prettier (for code compilation).
- B Mocha Test Explorer (to run and debug test files visually) and Hardhat (for Solidity language support and integrations for Hardhat projects).**
- C GitLens (for version control) and Docker (for containerization).

Explain the purpose of a local test network provided by Hardhat, as shown when running the `'npx hardhat accounts'` task.



- A It provides pre-funded real-world Ethereum accounts for immediate deployment to the mainnet.
- B It simulates a blockchain environment on the developer's local machine, offering a list of pre-funded accounts for quick and isolated testing and scripting without interacting with public testnets.**
- C Its primary function is to serve as a decentralized storage solution for dApp data.

Why does the lecturer strongly advise against using AI-generated code for smart contracts?



- A AI-generated code is always more complex and harder to understand.
- B AI-generated code cannot be audited or maintained by humans.
- C Smart contracts require absolute clarity, auditability, and battle-tested security, which AI-generated code currently lacks.